

M6 - Code Representation

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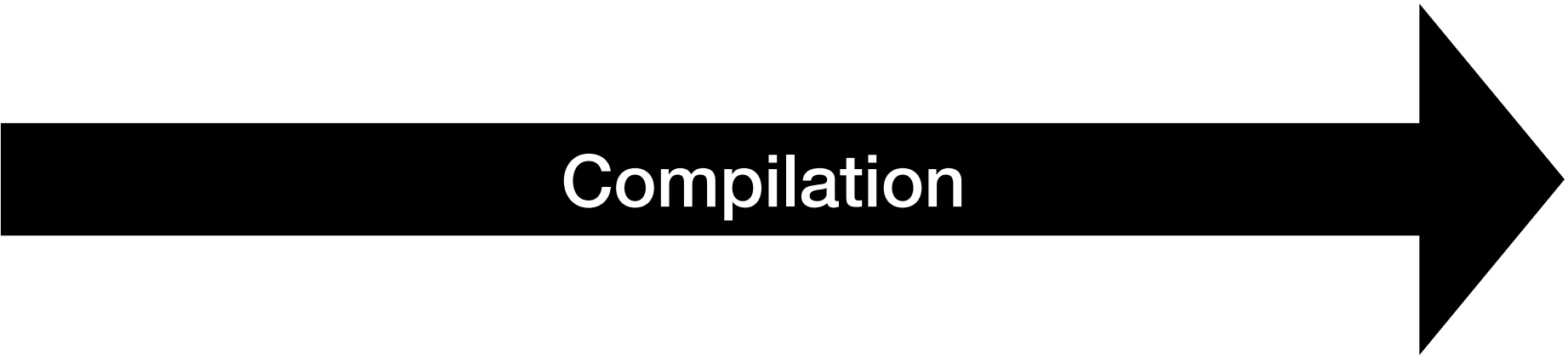
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Goals

- Understand Code Representation
 - Different Intermediate representations
 - Translations
 - Analyses
 - Compiler-based examples

Compilers

```
MyClass >> foo  
  ^ 1 + 17
```



Compilation

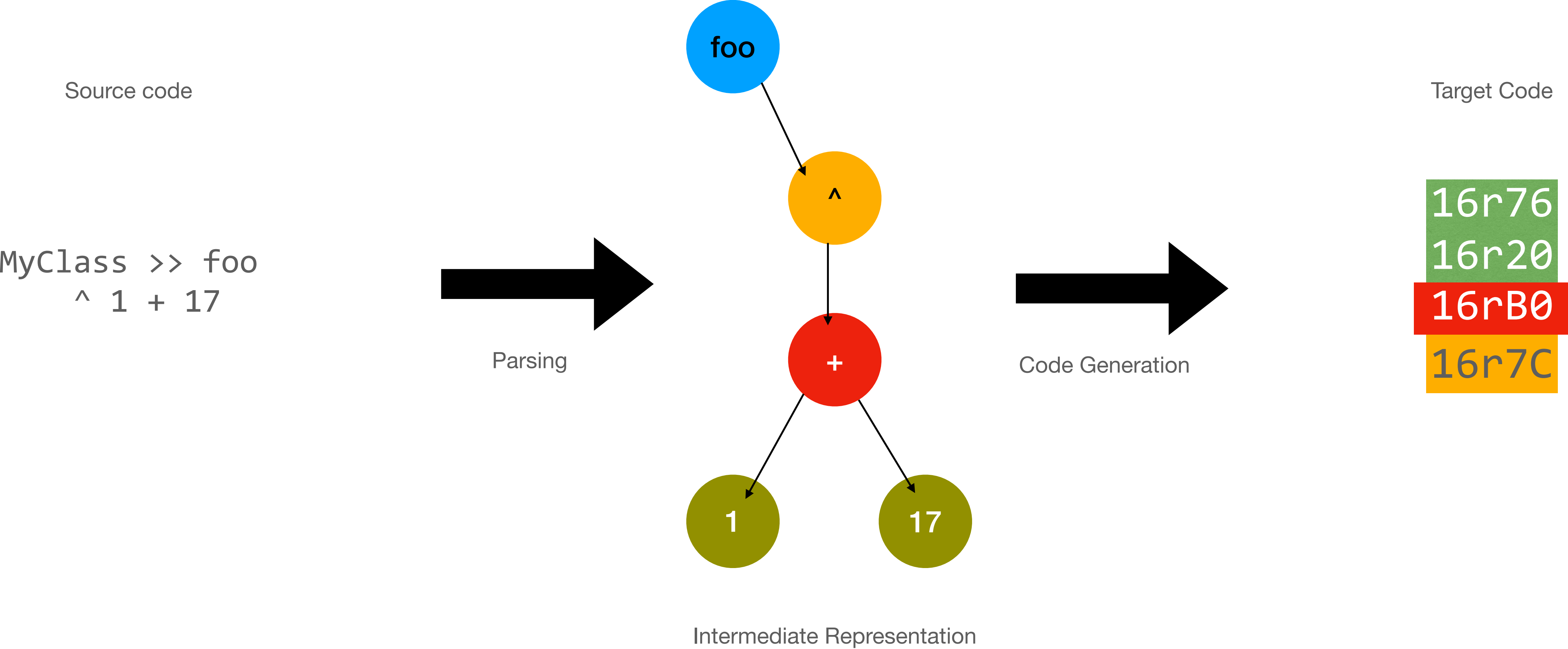
```
push 1  
push 17  
send +  
returnTop
```

A program that translates a program in a *source* language to a *target* language

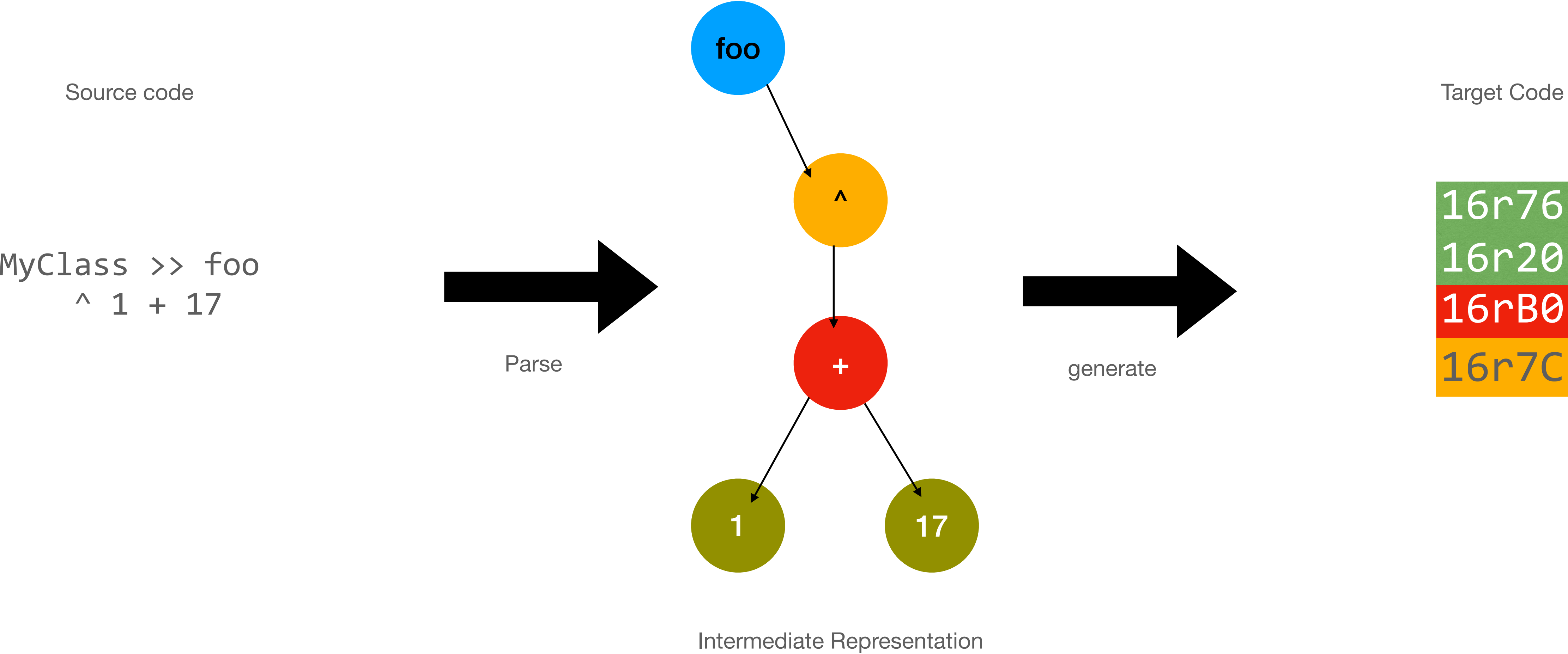
Target Code

- Another programming language => we talk about transpilation
e.g., Pharo to C translation
- Some binary code for a virtual machine
e.g., the Pharo bytecode
- Some binary code for a real machine
e.g., machine code for x86, or ARMv8

Compiler Internals Example: Simple Compiler

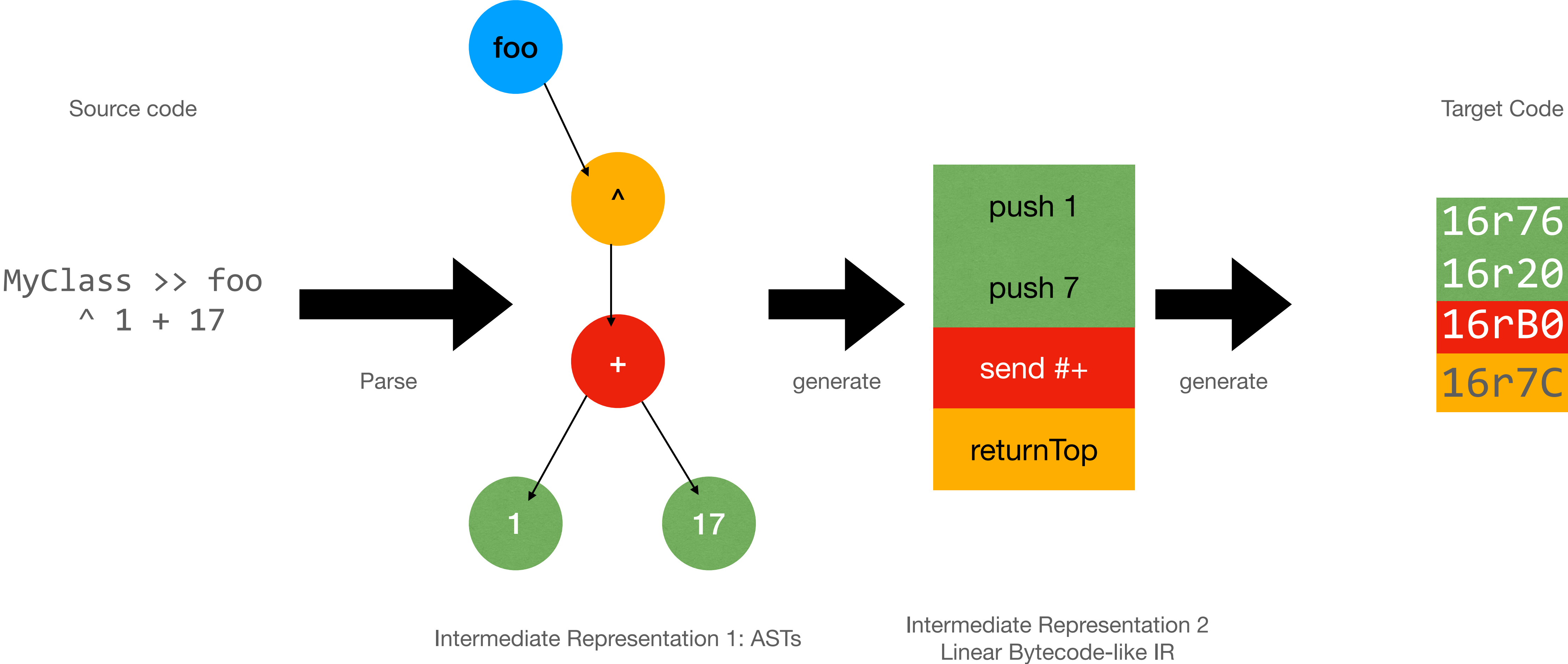


Real Example: The old Pharo Compiler

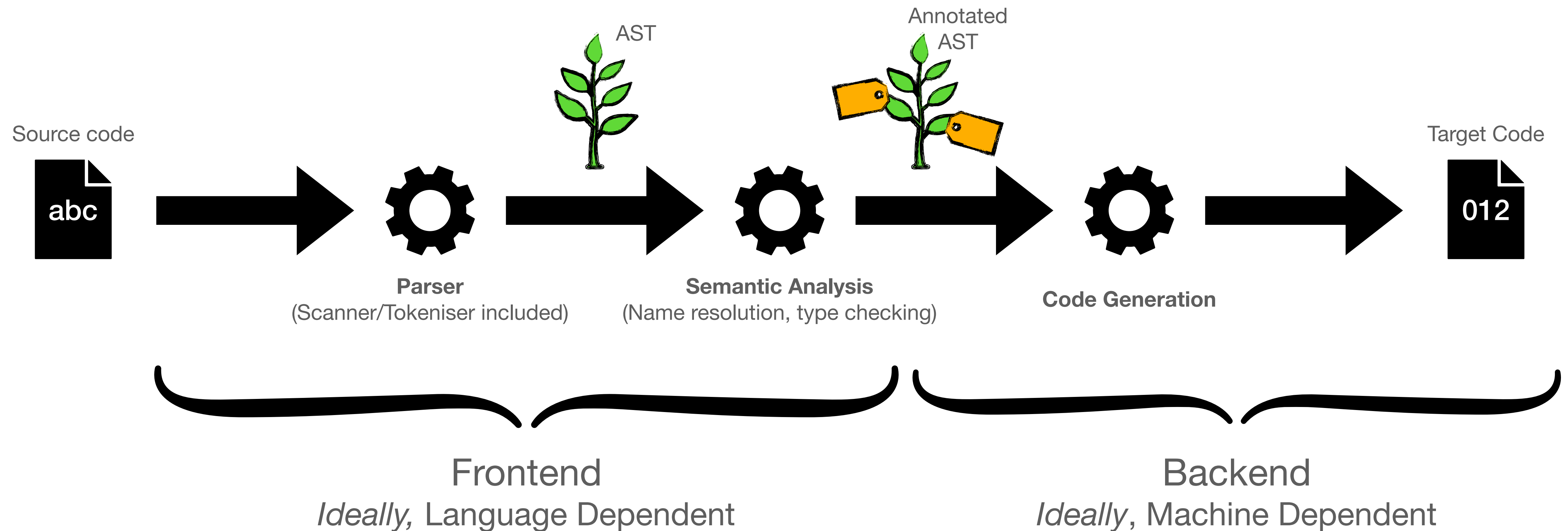


Real Example: The Pharo Opal Compiler

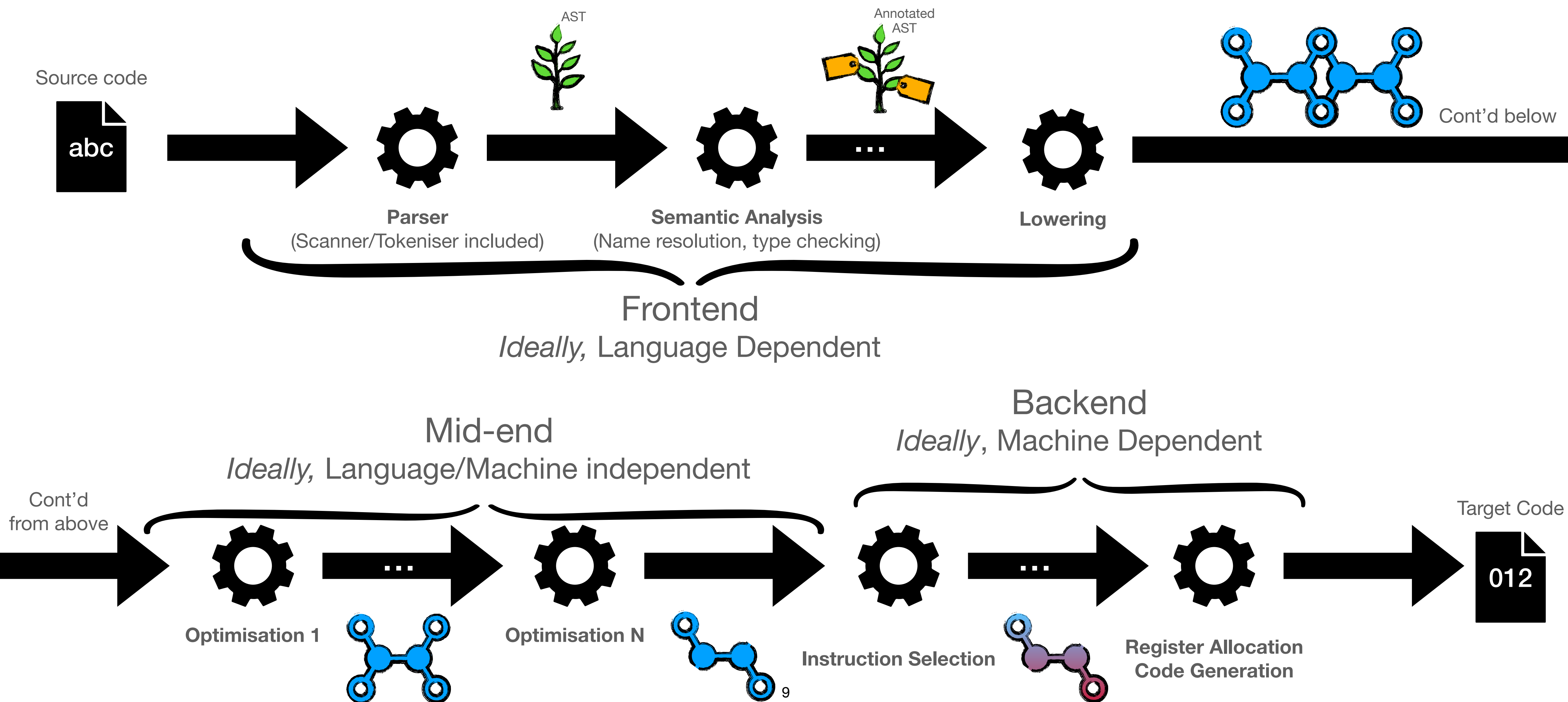
Introducing linear intermediate representations



Compiler Phases by Example

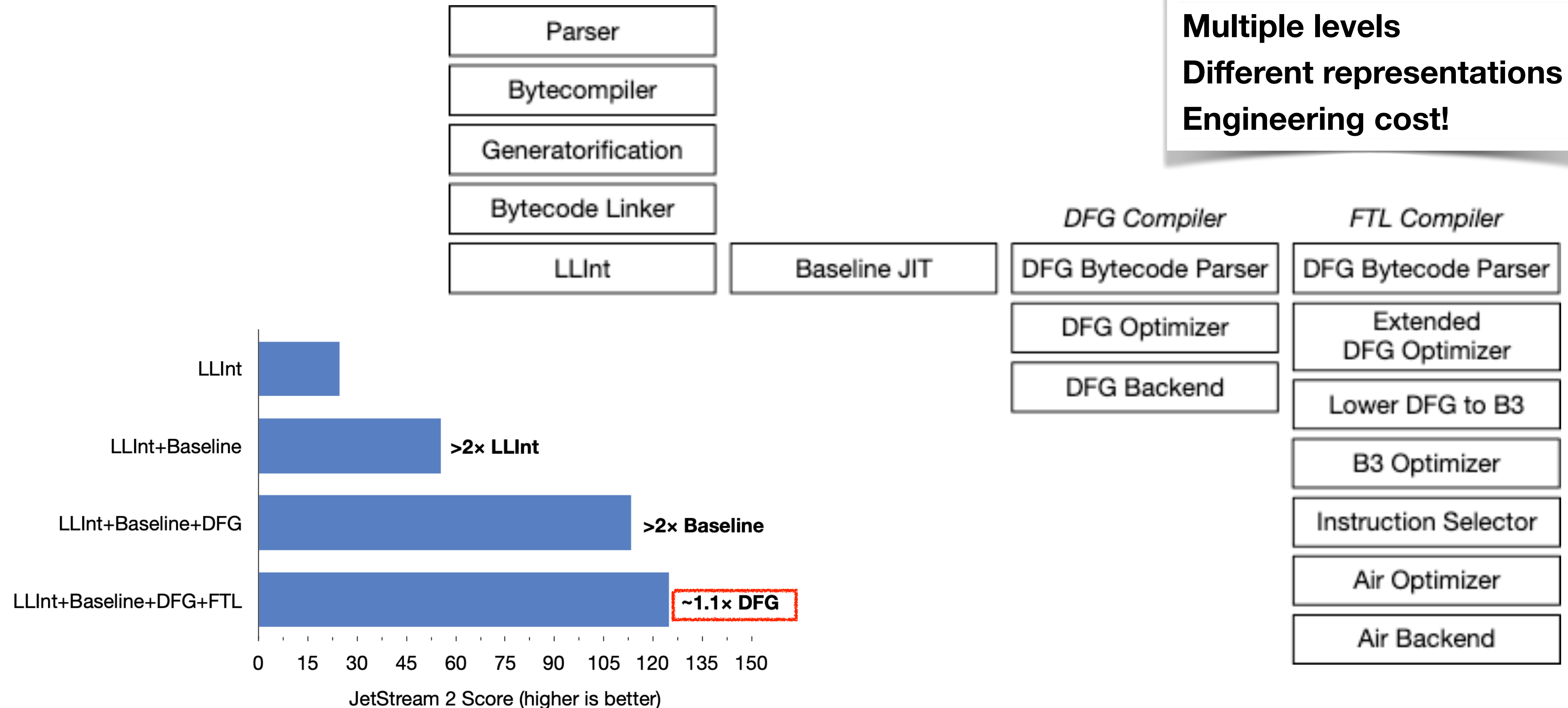


Modern Optimizing Compiler Phases



Javascript Core JIT Compiler

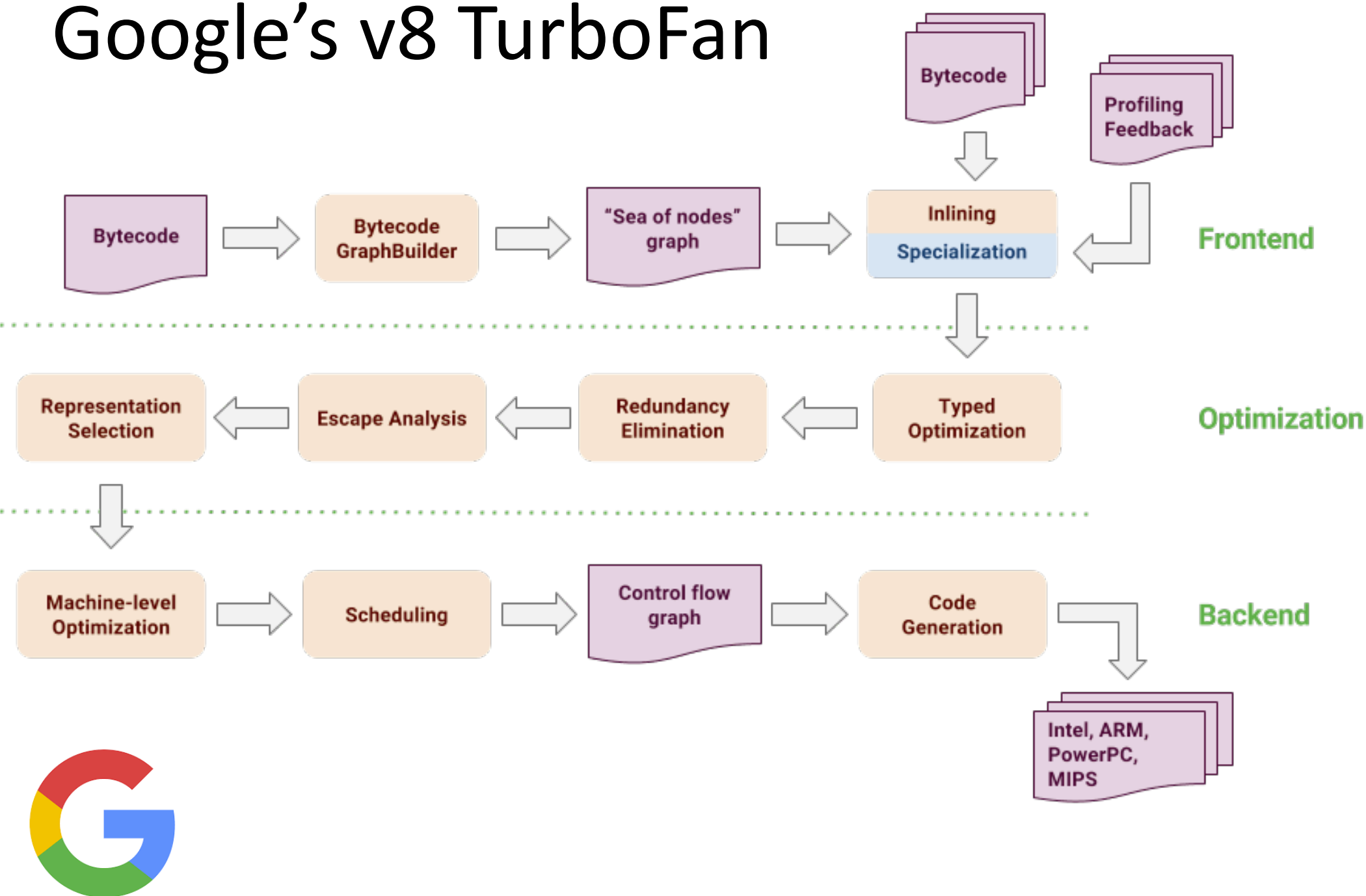
Multiple levels
Different representations
Engineering cost!



V8 and JSCore in depth

Apple's JavascriptCore[2021]

Google's v8 TurboFan



<https://webkit.org/blog/10308/speculation-in-javascriptcore/>
<https://ponyfoo.com/articles/an-introduction-to-speculative-optimization-in-v8>



Intermediate Representation design:

Notes and Terminology

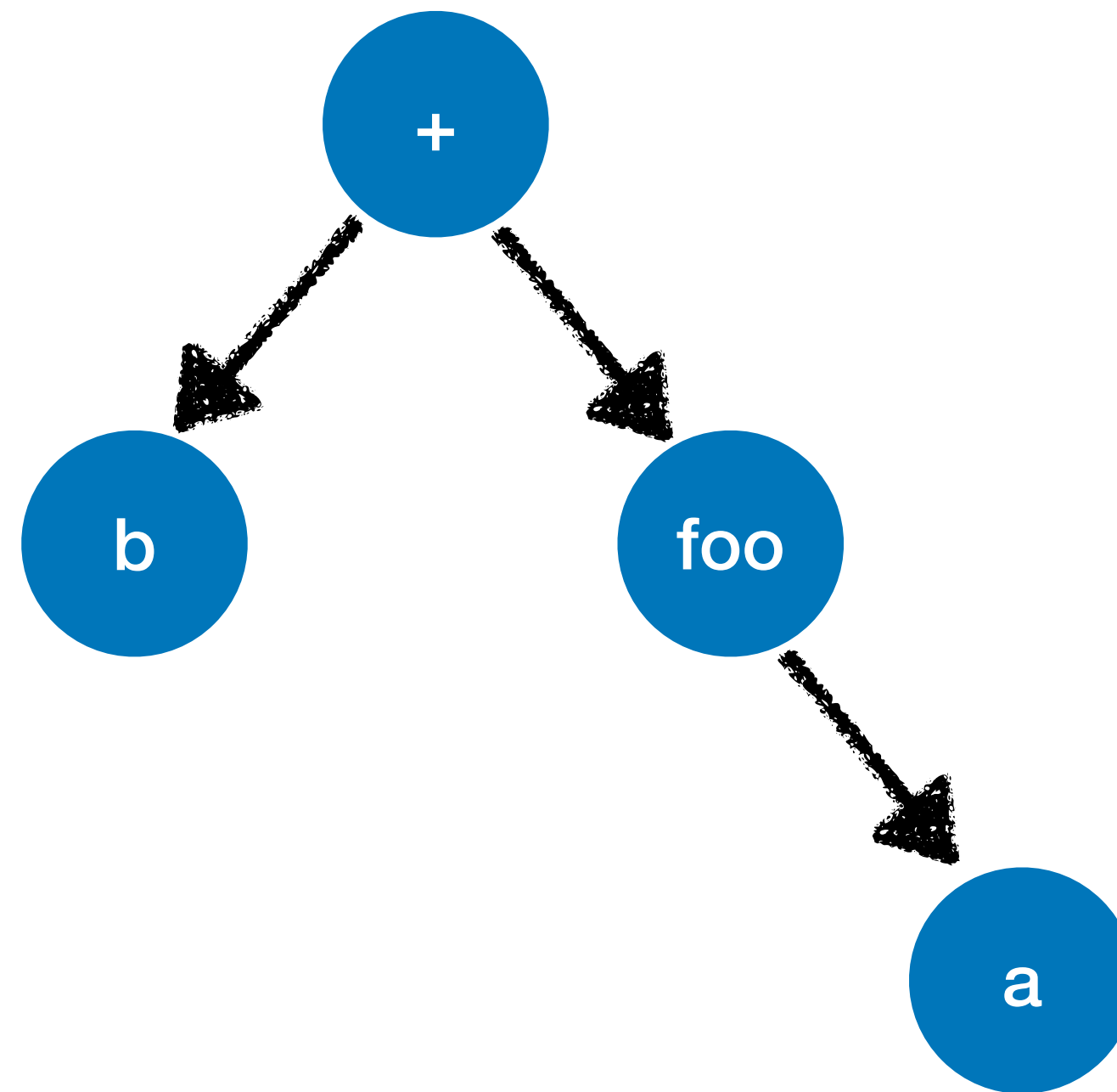
IRs: Data structures to represent code

`b + foo(a);`

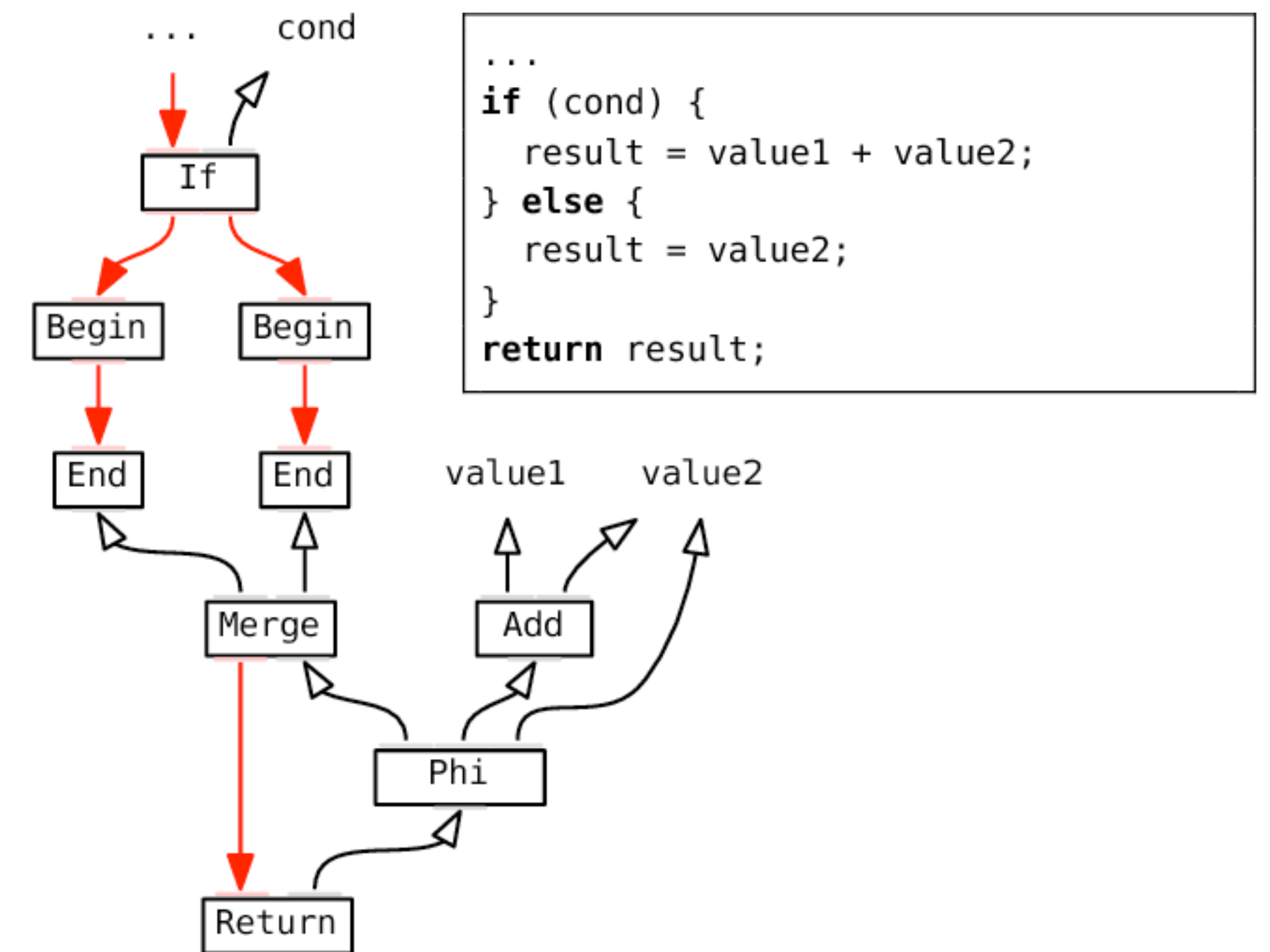
- Lists

LOAD R1, b
MOV R2, R1
LOAD R1, a
CALL FOO
ADD R1, R1, b

- Trees



- DAGs



Lists representing code

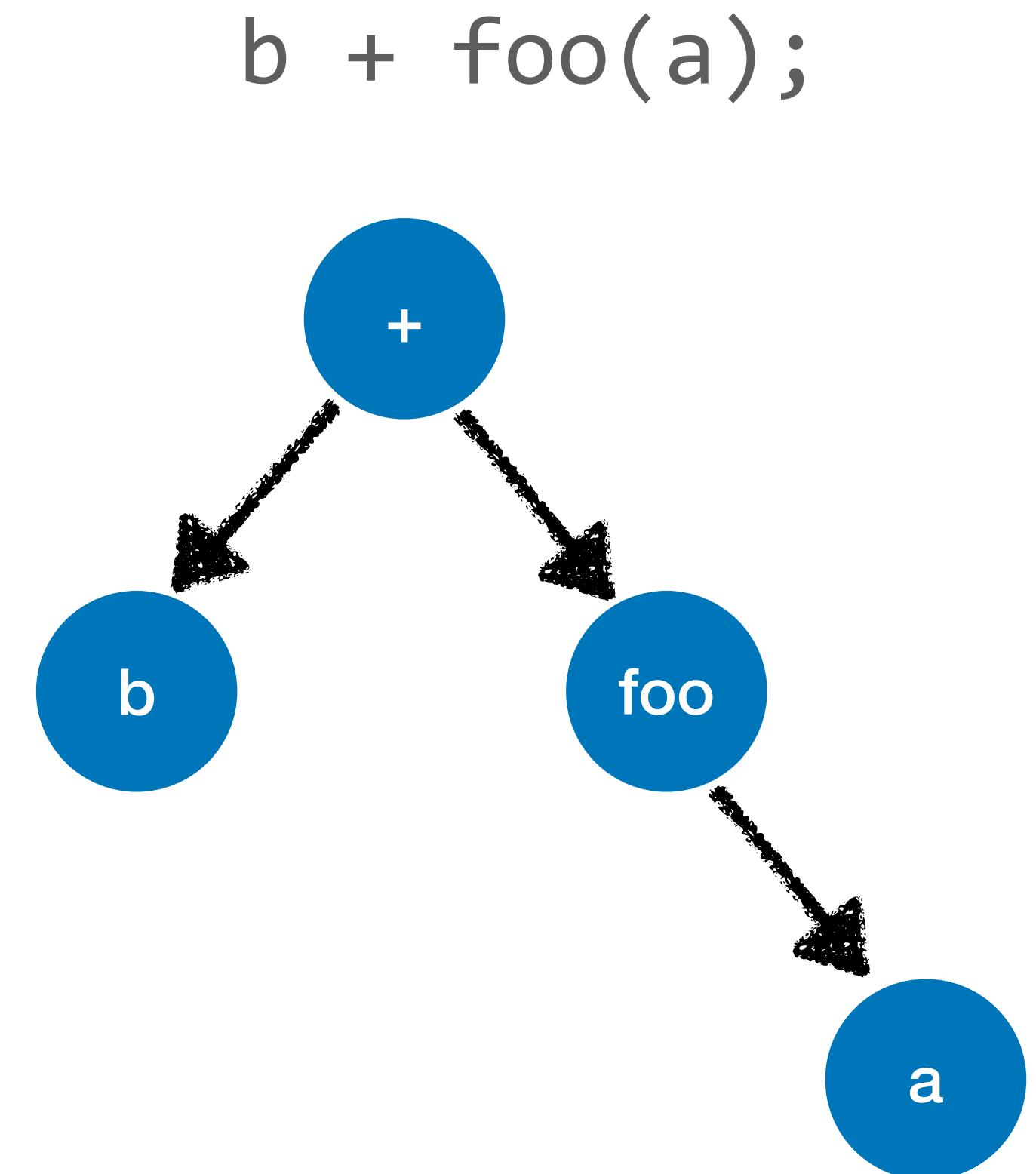
- Closer to “machine” code
- Simple to manipulate
- Relations between instructions become implicit
 - e.g., how many arguments does foo have?
 - e.g., Answer => sometimes, we need to see foo’s code
 - These become “conventions”

`b + foo(a);`

LOAD R1, b
MOV R2, R1
LOAD R1, a
CALL foo
ADD R1, R1, b

Trees representing code

- Closer to source code
- Often produced by a parser
- Relations are explicit
 - e.g., how many arguments does foo have?
 - e.g., Answer => look at foo's children!



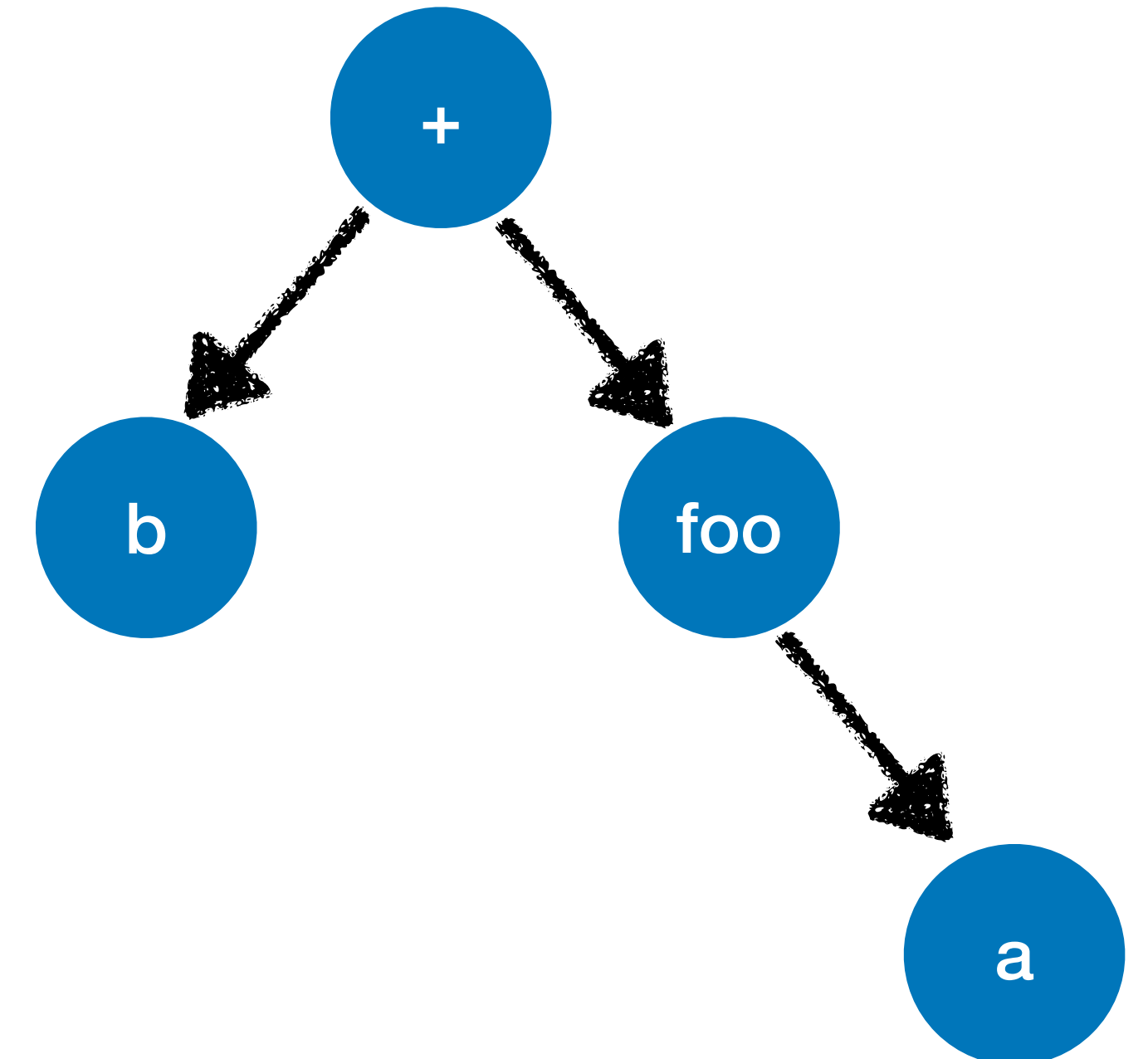
Abstract Syntax Trees (ASTs)

- Trees representing code
- Abstract, because they do not represent ALL elements in the grammar
 - i.e., parentheses, statement finalisers, indentation are **not** in the tree

b + foo(a);

b + (foo(a));

b +
foo(a)



Other Possible IR Flavors

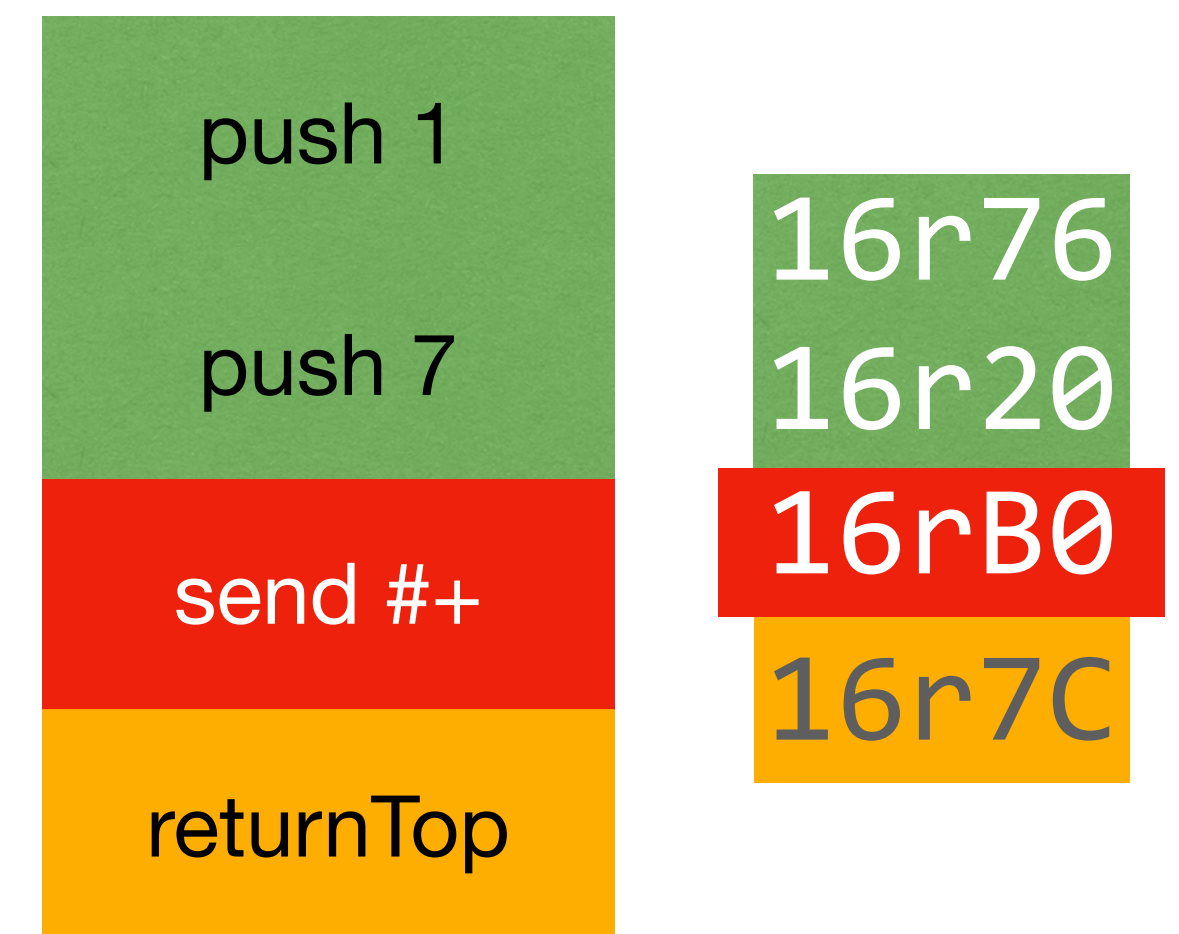
- Control Flow Graphs
 - Nodes represent uninterrupted sequences of instructions
 - Edges represent control-flow
- Sea of nodes
 - Flat representation: control flow and instructions are at the same level
- Single Static Assignment (SSA) form + variants
 - Each variable has max one assignment

On Code Generation:

Notes and Terminology

Target code: virtual "machine code"

- A virtual machine simulates a machine
- The instructions are called **bytecodes**
- Independent of the real machine
- Stack based: operands are exchanged through a stack
- Compact: Stack is implicit



Target code: machine code

- Binary code executed by the machine/CPU/GPU/...
- Bytes encode instructions
- The set of instructions + CPU is called **instruction set architecture (ISA)**
- Each machine/CPU has its own ISA and binary encoding of it
- Typically register machines:
operands are exchanged through registers and memory

#[31 32 3 213 76 1 0 88]

some arm v8 instructions

Machine code VS Assembly code

- Assembly is a programming language
- (not machine code)
- That is translated to machine code using a compiler (aka an assembler)

```
nop  
ldr x12, #40
```

```
#[ 31 32 3 213 76 1 0 88 ]
```

```
nop
```

```
ldr x12, #40
```

For simplicity: we will use assembly examples to represent machine code

Machine code instructions

- Basic instructions:

mnemonic op1 op2 op3 ...

- Write data to memory
- Read data from memory
- Arithmetic (+ , - , * ...)
- Bit instructions (shift, bitAnd ...)
- Control flow (jump, jump if)

e.g.,
load r1 [r1, #40]
add r3 r2 r1
store [r1, #40] r3

Machine code: Instruction Operands

Basic operand types

- Immediate numbers: encoded directly in the instruction bytes

e.g., `#40`

```
load r2 [r1, #40]
```

- Registers: small memories in the CPU

```
add r3 r2 r1
```

```
store [r1, #40] r3
```

e.g., `r1 r2 r3`

- Memory addresses: calculated from registers and/or immediates

e.g., `[r1, #40]`

Machine code: Registers

- Small memories in the CPU
- Some are for general use
- Some are specific (e.g., Floating point numbers)
- Some are for the CPU (and not for us)

=> And there are very few!!!

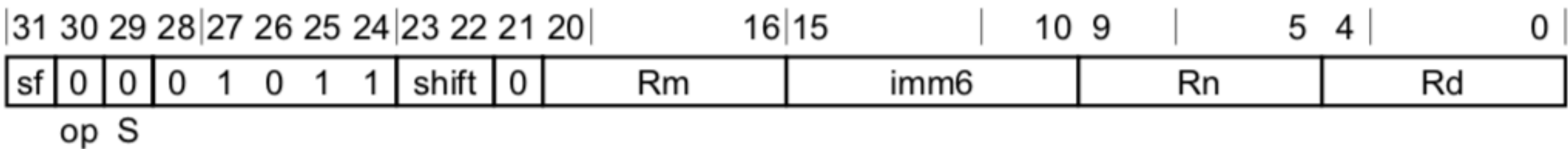
e.g.,
load r1 [r1, #40]
add r3 r2 r1
store [r1, #40] r3

Machine code: binary encoding

Each ISA has its manual

C6.2.5 ADD (shifted register)

Add (shifted register) adds a register value and an optionally-shifted register value, and writes the result to the destination register.



32-bit variant

Applies when sf == 0.

ADD <Wd>, <Wn>, <Wm>{, <shift> #<amount>}

64-bit variant

Applies when sf == 1.

ADD <Xd>, <Xn>, <Xm>{, <shift> #<amount>}

Takeaways

- Compilers are organized in Phases
- Compilers represent code using Intermediate Representations (IRs)
- Compilers are designed with different IRs to achieve different trade-offs
- Phases either
 - modify some code representation e.g., optimizations
 - change some code representation into another e.g., a parser

Material

- Books
 - Dragon Book
 - Advanced Compiler Design & Impl
 - Engineering a Compiler
- Blogs
 - V8, JSCore

